State Vaccination Policies, Delayed Care, and Health Expenditures

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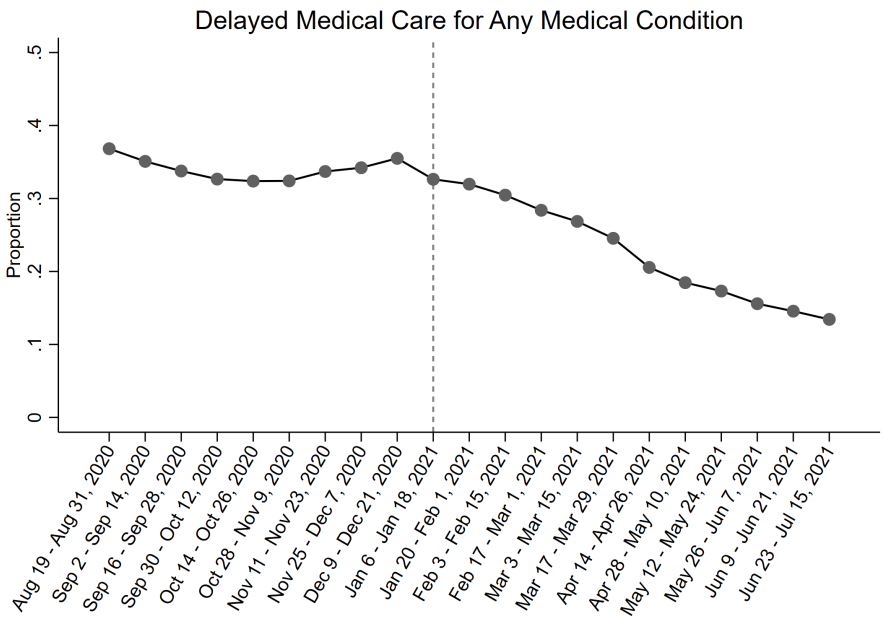
Motivation: delayed or skipped care

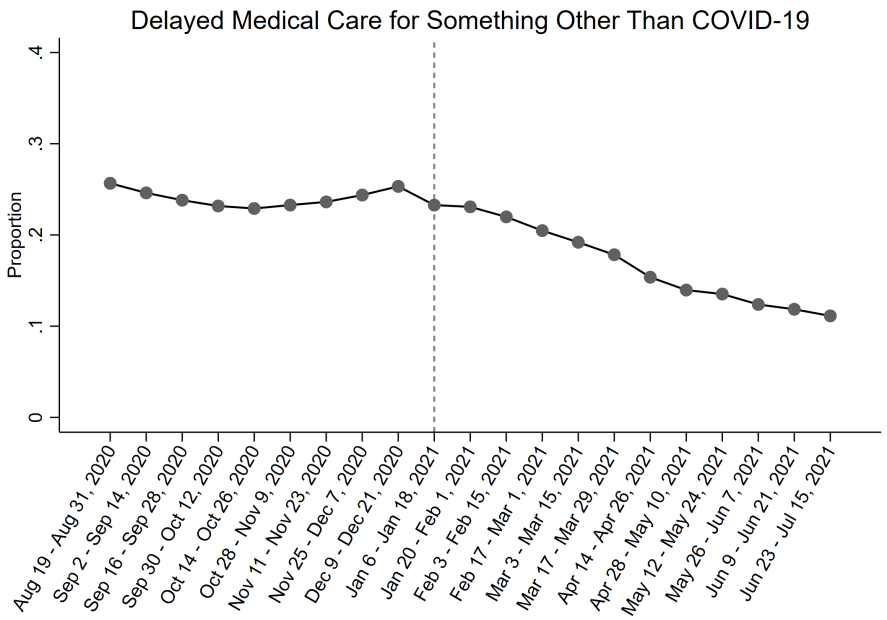
- More than **30%** of individuals delayed or skipped care during COVID-19.
  - Concerns about virus exposure.
- Increase morbidity and mortality risk associated with treatable and preventable health conditions.
- Contribute to reported excess deaths directly or indirectly related to the pandemic.
- Ramifications are more pronounced for children.
  - Spillover to outcomes other than health.
  - Educational outcomes/human capital formation (Cohodes et al., 2016; Miller and Wherry, 2019; Brown, Kowalski, and Lurie, 2020).
  - Criminal outcomes (Arenberg, Neller, and Stripling, 2020).

Motivation: vaccine rollout

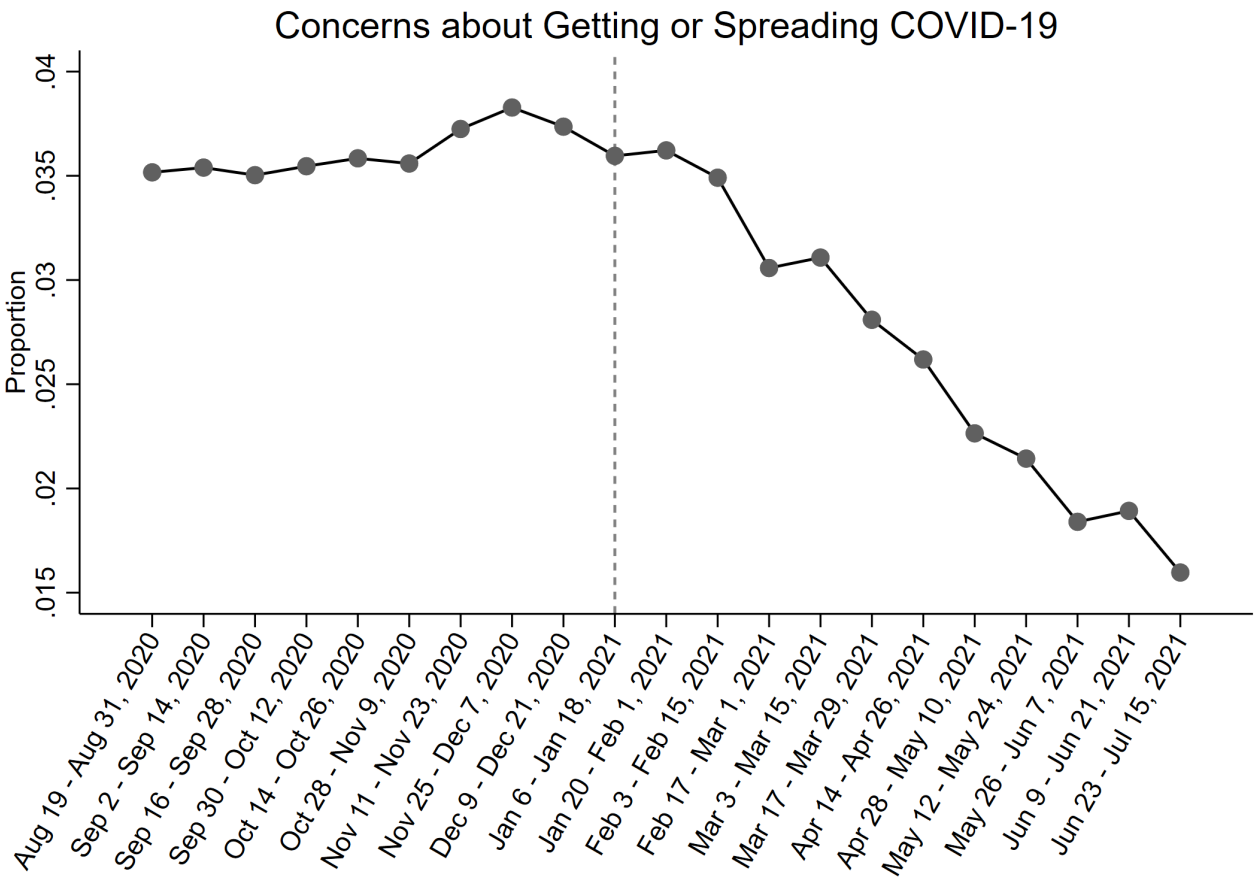
- Medical innovations - development of vaccines.
- Accelerated approval of vaccines (e.g., COVID-19 vaccines).
- Mitigate concerns about exposure to coronavirus.
- Reduce tendencies to delay or put off care.
- Well, is that the case?
    - Our paper shows that the answer is affirmative.

Descriptive evidence: delayed care





Descriptive evidence: concerns/worries



What we do

- Evaluate the causal effect of COVID-19 vaccination on the likelihood of delaying care.
  - any medical conditions
  - medical conditions other than COVID-19
- Address selection into receiving vaccination by exploiting the arguably exogenous variation in age-specific vaccine eligibility rollout.
  - Agrawal et al. (2021) construct a similar instrument
  - investigate the impact of COVID-19 vaccination on mental health outcomes
- Investigate (positive) spillovers to children in the household.
- Explore whether vaccination has any effect on consumer healthcare spending in the short run.

What we find

- Reduction in the likelihood of delaying care.
  - any medical conditions (17.8%)
  - medical conditions other than COVID-19 (32.4%)
- Salient effects on:
  - minorities
  - and those with lower socioeconomic status
- No evidence of spillovers to children.
- Positive but statistically insignificant effects on health expenditures.

Contributions

- First evidence on the causal effect of COVID-19 vaccination on delaying medical care or avoidance.
  - Earlier experimental evidence: vaccine anticipation and social distancing behavior (Andersson et al., 2021).
  - Observational evidence: mobility (Zhong et al., 2021), mental health outcomes (Agrawal et al., 2021), mortality (Gupta et al., 2021; Barro, 2022).
- Contribute to the evidence on vaccination and the COVID-19 outbreak:
  - vaccine hesitancy/vaccine uptake (Khubchandani and Macias, 2021; Momplaisir et al., 2021; Karaivanov et al., 2021)
  - healthcare utilization (Giannouchos et al., 2021; Cantor et al., 2022)
  - economic activity (Chetty et al., 2020)

Contributions

- Shed light on mechanisms:
  - reducing concerns about getting or spreading COVID-19
- Have important policy implications:
  - vaccines have the effect of not only protect against coronavirus, but also safeguard against the worsening of health due to delayed care
  - medical innovations might in general improve public health by influencing people's assessment of risk during a pandemic

Data and analysis sample

- Individual vaccine uptake, delaying medical care, & concerns.
- Household Pulse Survey (HPS), US Census Bureau.
  - A 20-minute online survey with multiple waves including adults aged 18 and older
  - Employ survey waves 22 (Jan 6, 2021) - 33 (July 5, 2021)
  - Questions on COVID-19 vaccines were added in survey wave 22
  - About 700,000 observations for delayed care and vaccination status
  - Survey questions on vaccine uptake and delaying medical care Survey Questions
  - Descriptive statistics Descriptive Statistics

Empirical methodology

- We formalize our research design using an IV specification:

$$Y_{ist} = \beta_0 + \beta_1 V_{ist} + \beta_2 X_{ist} + \delta_s + \gamma_t + \epsilon_{ist}, \quad (1)$$
$$V_{ist} = \alpha_0 + \alpha_1 Z_{ist} + \alpha_2 X_{ist} + \delta_s + \gamma_t + \nu_{ist}, \quad (2)$$

where

$$\mathbb{E}[\epsilon_{ist}, \nu_{ist} | X_{ist}, \gamma_t, \delta_s] \neq 0, \mathbb{E}[\epsilon_{ist}, Z_{st} | X_{ist}, \gamma_t, \delta_s] = 0, \text{ and } \alpha_1 \neq 0.$$

  - Y_{ist} is an indicator for delayed care for individual i in state s at time (survey wave) t
  - V_{ist} is receiving COVID-19 vaccine
  - Z_{ist} is age-specific vaccine eligibility
  - X_{ist} is observable characteristics (gender, race/ethnicity, marital status, education, household income, and health insurance)
  - δ_s and γ_t are state and survey wave fixed effects
  - Standard errors are clustered at the state level

Instrument validity: independence

- Independence (or exogeneity) assumption requires our IV to be as good as random.
- Age-specific vaccine eligibility rollout has to be independent of the outcome variable and the potential treatment assignment.
- Sufficient for a causal interpretation of the reduced form
- We tackle this problem in two ways:
  - Run a dynamic DID using vaccination status and delayed care
  - Check for pre-trends
  - Test for a (reverse) feedback effect running from delayed care to vaccine eligibility
- Another alternative: balancing test (see if the IV is correlated with observable characteristics)
  - Doesn't always work. What if you have *conditional* independence!

Instrument validity: exclusion restriction

- Exogeneity assumption does not ensure that the exclusion restriction is satisfied.
- Exclusion restriction states that vaccine eligibility can only affect delayed care through its impact on vaccination status.
- There could be alternative paths through which vaccine eligibility influences the likelihood of delaying care.
  - Let's call these alternative paths "**backdoor paths**"
  - Simple illustration of backdoor paths

DAG

Instrument validity: exclusion restriction

- Two potential backdoor paths: $Z_{ist} \leftarrow P \rightarrow Y_{ist}$ and $Z_{ist} \rightarrow S \rightarrow Y_{ist}$.
- P: pre-existing economic conditions, health measures, and policy responses to the pandemic
- S: spillovers to unvaccinated individuals
  - Relaxed clinical guidelines (e.g., phone screening): lowers the opportunity cost of healthcare access
  - Vaccine eligibility rollout can make these individuals feel safer
  - Spillovers within social networks: positive or negative messages (or posts) from eligible individuals
- Condition on P and S to close these backdoor paths.
  - P: (lagged) COVID-19 death rates and the stringency index; and (lagged) unemployment rate
  - S: region-by-month fixed effects, state-specific linear trends, or state-by-wave fixed effects
  - More on S: (lagged) friend exposure to vaccination information using the Social Connectedness Index from Facebook and data on vaccination rates.

Instrument validity: relevance

- Age-specific vaccine eligibility should have a strong effect on vaccination status.
- We conduct a range of tests for underindentification and weak instruments.
- We have a relatively strong instrument in this setting.
- Kleibergen-Paap rank Wald F-statistic ranges from 142.2 to 185.1.
- Our first-stage results complement existing studies that show a substantial increase in vaccination uptake in Canada and Europe (i.e., France, Italy, and Germany) due to COVID-19 vaccination mandates (Karaivanov et al., 2021).

Instrument validity: monotonicity

- Monotonicity requires that

$$\mathbb{E}[V_{ist}|X_{ist}, \gamma_t, \delta_s, Z_{ist} = 1] \geq \mathbb{E}[V_{ist}|X_{ist}, \gamma_t, \delta_s, Z_{ist} = 0] \textbf{ or}$$
$$\mathbb{E}[V_{ist}|X_{ist}, \gamma_t, \delta_s, Z_{ist} = 1] \leq \mathbb{E}[V_{ist}|X_{ist}, \gamma_t, \delta_s, Z_{ist} = 0]$$

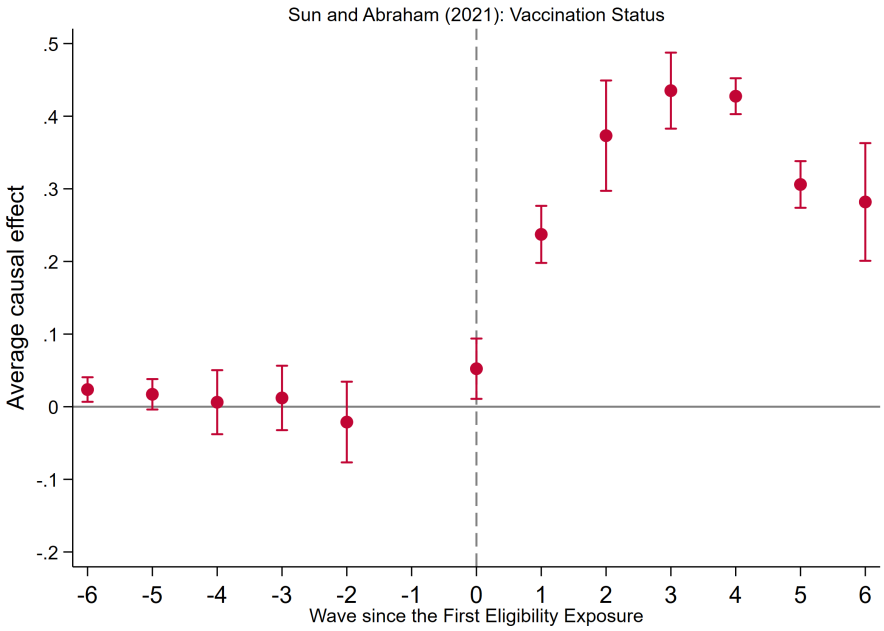
- Age-specific vaccine eligibility weakly operates in the same direction on all individuals.
- A failure of monotonicity means age-specific vaccine eligibility pushes some people into treatment while pushing others out ("**defiers**").
- A potential issue in leniency designs (Frandsen, Lefgren and Leslie, 2019). Ex ante this issue does not seem to be prevalent in our setting.
- We nonetheless present first stage results for different subsamples and consistently find a positive relationship.

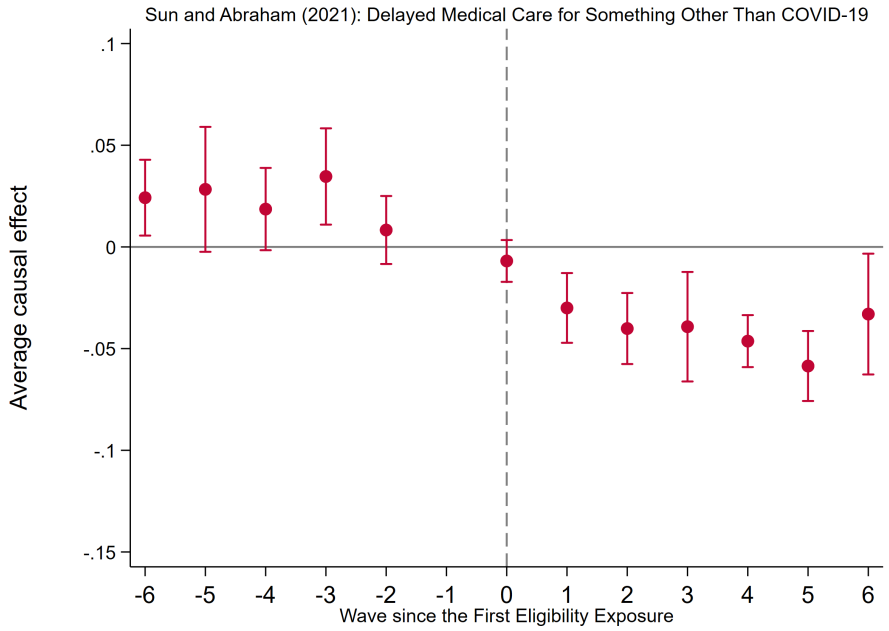
Reduce form: vaccine eligibility and delayed care

	(1)	(2)	(3)	(4)	(5)
Panel A: Delayed Medical Care for Any Medical Condition					
Vaccine Eligibility	-0.0094**	-0.0101**	-0.0101**	-0.0101**	-0.0102**
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
Panel B: Delayed Medical Care for Something Other Than COVID-19					
Vaccine Eligibility	-0.0120***	-0.0123***	-0.0123***	-0.0123***	-0.0123***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)
N	700,070	700,070	700,070	700,070	700,070
State FE	√	√	√	√	√
Survey Wave FE	√	√	√	√	√
Region × Wave FE		√	√	√	√
Health measures and policy responses			√	√	√
Economic conditions				√	√
Social networks					√

Reduce form: imputation estimators

- Treatment heterogeneity: biased ATTs. We use the Sun and Abraham estimator (Sun and Abraham, 2021, *Journal of Econometrics*)
- We do not condition on any covariates
- Similar findings for any medical condition





OLS

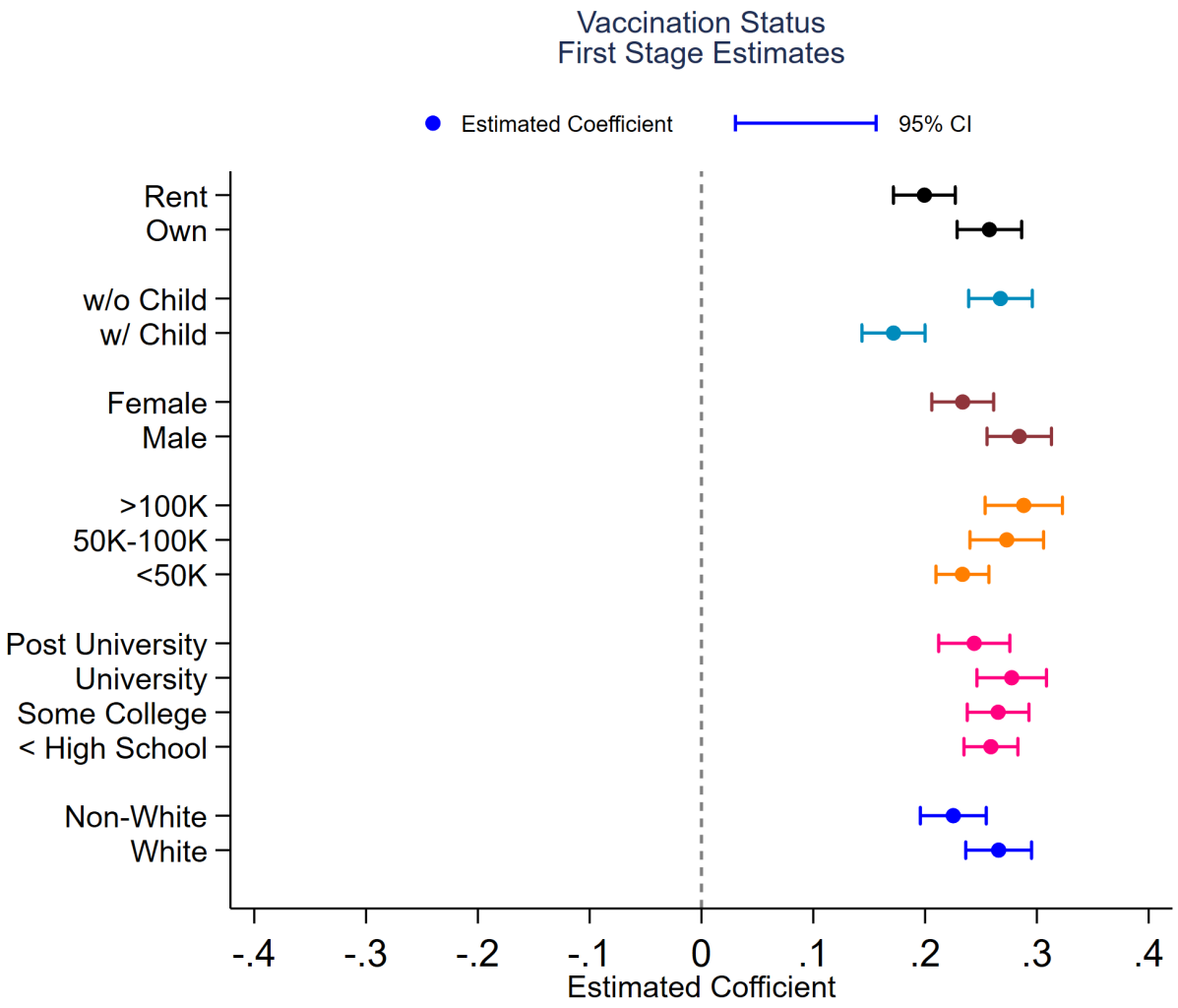
Baseline results: vaccination status and delayed care

- Column (6): any medical condition (17.8%); conditions other than COVID-19 (32.4%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
			OLS					IV		
Panel A: Delayed Medical Care for Any Medical Condition										
Received Vaccine	-0.0159***	-0.0159***	-0.0159***	-0.0159***	-0.0159***	-0.0606**	-0.0634**	-0.0635**	-0.0636**	-0.0640**
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)
Panel B: Delayed Medical Care for Something Other Than COVID-19										
Received Vaccine	-0.0216***	-0.0217***	-0.0217***	-0.0217***	-0.0217***	-0.0775***	-0.0770***	-0.0771***	-0.0771***	-0.0772***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.023)	(0.025)	(0.025)	(0.025)	(0.025)
						First Stage Estimates (*Received Vaccine*)				
Vaccine Eligibility						0.1554***	0.1592***	0.1592***	0.1591***	0.1588***
						(0.011)	(0.013)	(0.013)	(0.013)	(0.013)
N	700,070	700,070	700,070	700,070	700,070	700,070	700,070	700,070	700,070	700,070
State FE	√	√	√	√	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√	√	√	√	√
Region × Wave FE		√	√	√	√		√	√	√	√
Health measures and policy responses			√	√	√			√	√	√
Economic conditions				√	√				√	√
Social networks					√					√
Kleibergen-Paap rk LM statistic						18.03	17.69	17.67	17.64	17.59
Kleibergen-Paap rk Wald F statistic						185.1	145.8	145.1	143.8	142.2

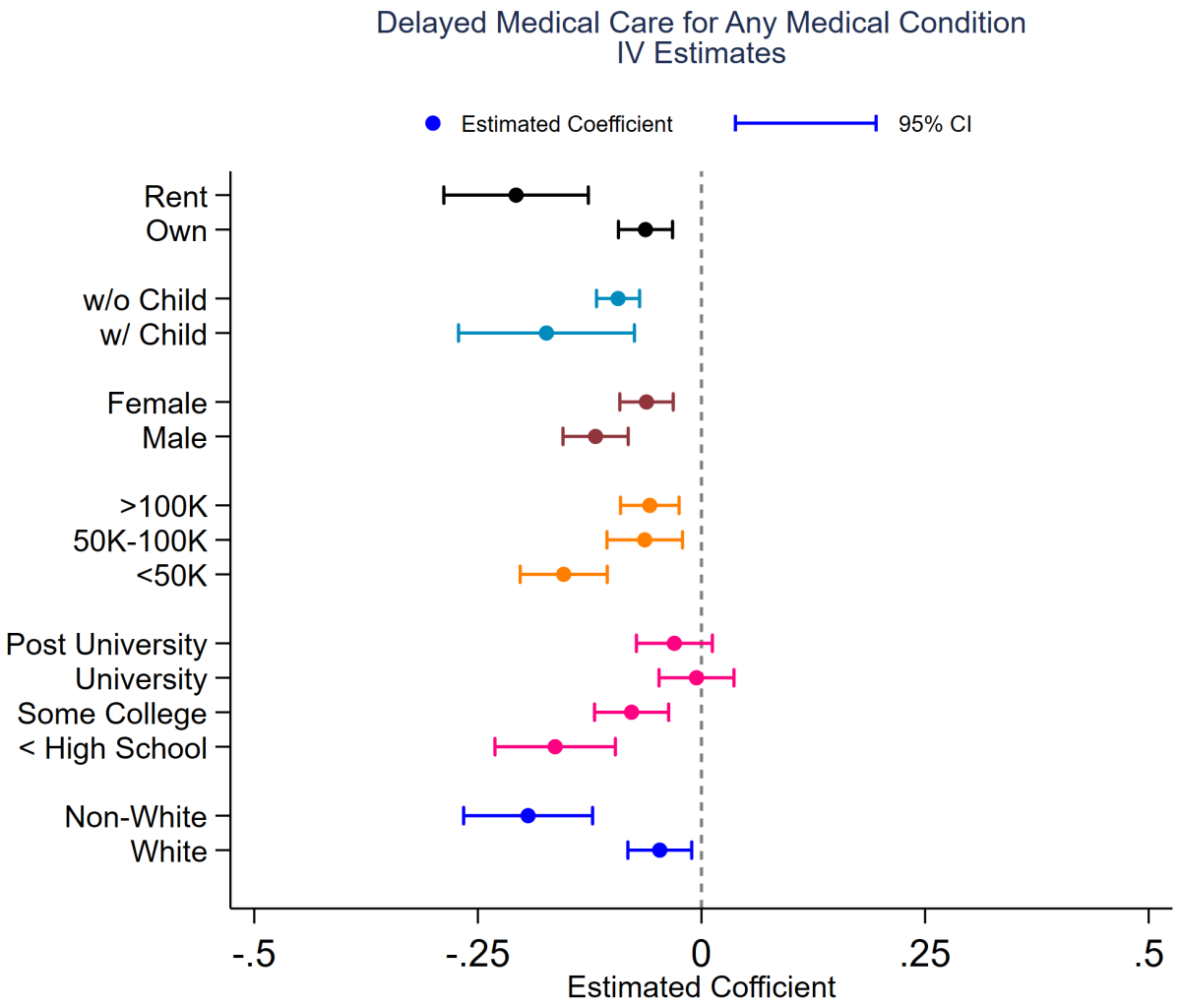
Heterogeneity: first stage

- Cannot reject instrument monotonicity



Heterogeneity: IV estimates

- Similar findings for medical conditions other than COVID-19



Robustness Checks

- Trend analysis (region-specific linear trends, state-specific linear trends, state-by-wave FE) Trend Analysis
- Testing for feedback effects Feedback Effects
- Adjusting policy assignment Policy Assignment
- Saturated models and excluding controls (in progress)
  - LATE interpretation does not necessarily hold when the IV specification includes covariates
  - Decomposition of the TSLS estimand with covariates shows that it not only reflects the treatment effects for **compliers**, but also the (negatively weighted) **always-takers** (Blandhol et al., 2022).
  - Non-parametrically control for covariates.

Mechanisms: fear of getting or spreading COVID-19

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
			OLS					IV		
Panel A: Concerned about Getting or Spreading COVID-19										
Received Vaccine	0.0037	0.0037	0.0037	0.0037	0.0037	-0.0288**	-0.0296**	-0.0296**	-0.0294**	-0.0295**
	(0.002)	(0.003)	(0.003)	(0.003)	(0.002)	(0.011)	(0.012)	(0.012)	(0.012)	(0.012)
						First Stage Estimates (*Received Vaccine*)				
Vaccine Eligibility						0.1857***	0.1873***	0.1873***	0.1871***	0.1869***
						(0.014)	(0.015)	(0.015)	(0.015)	(0.015)
N	288,429	288,429	288,429	288,429	288,429	288,429	288,429	288,429	2288,429	288,429
State FE	√	√	√	√	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√	√	√	√	√
Region × Wave FE		√	√	√	√		√	√	√	√
Health measures and policy responses			√	√	√			√	√	√
Economic conditions				√	√				√	√
Social networks					√					√
Kleibergen-Paap rk LM statistic						16.71	16.86	16.86	16.83	16.83
Kleibergen-Paap rk Wald F statistic						164.9	157.4	158.3	157	156.3

No spillovers to children

- Parents' vaccination is **necessary**, but not **sufficient** to ease concerns to delay or avoid care for their children

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
			OLS					IV		
Panel A: Children Delayed Preventive Health Care										
Received Vaccine	0.0197**	0.0199**	0.0199**	0.0199**	0.0199**	-0.6080	-0.5064	-0.4996	-0.5206	-0.5827
	(0.009)	(0.009)	(0.009)	(0.009)	(0.009)	(0.541)	(0.387)	(0.383)	(0.350)	(0.353)
						First Stage Estimates (*Received Vaccine*)				
Vaccine Eligibility						0.0361**	0.0427***	0.0431***	0.0476***	0.0471***
						(0.014)	(0.013)	(0.014)	(0.015)	(0.015)
N	104,759	104,759	104,759	104,759	104,759	104,759	104,759	104,759	104,759	104,759
State FE	√	√	√	√	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√	√	√	√	√
Region × Wave FE		√	√	√	√		√	√	√	√
Health measures and policy responses			√	√	√			√	√	√
Economic conditions				√	√				√	√
Social networks					√					√
Kleibergen-Paap rk LM statistic						3.487	4.018	4.099	4.561	4.273
Kleibergen-Paap rk Wald F statistic						6.929	10.17	10.02	10.62	10.09

Vaccination rate and health expenditures

- Use novel (high frequency) credit and debit card spending data from Opportunity Insights (Chetty et al., 2020) - closely track the historical benchmarks of retail spending and services
- Vaccination status may be linked more broadly to the recovery of economic activities
  - null effect on health expenditures, but statistically significant effects on transportation spending

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
			OLS					IV		
Panel A: Consumer Health Care Spending										
Vaccination Rate per 100	0.0002	0.0007	0.0009	0.0006	0.0007	0.0014	0.0036	0.0031	0.0028	0.0044
	(0.0012)	(0.0018)	(0.0017)	(0.0013)	(0.0013)	(0.0034)	(0.0042)	(0.0042)	(0.0043)	(0.0043)
						First Stage Estimates (*Vaccination Rate per 100*)				
Fraction Eligible						0.0849***	0.0566***	0.0564***	0.0539***	0.0553***
						(0.0192)	(0.0132)	(0.0130)	(0.0122)	(0.0121)
N	8,750	8,750	8,750	8,750	8,050	8,750	8,750	8,750	8,750	8,050
State FE	√	√	√	√	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√	√	√	√	√
Region × Wave FE		√	√	√	√		√	√	√	√
Health measures and policy responses			√	√	√			√	√	√
Economic conditions				√	√				√	√
Social networks					√					√
Kleibergen-Paap rk LM statistic						11.13	9.89	10.09	10.54	10.46
Kleibergen-Paap rk Wald F statistic						19.55	18.40	18.88	19.41	20.90

Other Expenditure Categories

Conclusions

- Receiving vaccination reduces delayed care for any medical conditions and conditions other than COVID-19 by 17.8% and 32.4%, respectively.
- Stronger effects for Non-White individuals and those with lower socioeconomic status.
- Vaccination affects healthcare seeking behavior by reducing concerns about getting or spreading COVID-19.
- There are no (positive) spillovers to children.
- We find positive but insignificant effects on consumer healthcare spending.
  - positive and statistically significant effects on transportation and overall spending
- Our results highlight that medical innovations might influence people's assessment of risk during a pandemic.

Next steps

- Use a much larger sample size (more than **2.2 million** obs.): include waves 1-33.
- Check the robustness of the IV estimates using saturated models.
- Explore the reduce form effects using alternative DID estimators.
  - We already have some consistent results using the imputation estimators.

Appendix: Survey questions

- Delayed care for any medical condition:
  "At any time in the last 4 weeks, did you DELAY getting medical care because of the coronavirus pandemic?"
- Delayed care for conditions other than COVID-19:
  "At any time in the last 4 weeks, did you need medical care for something other than coronavirus, but DID NOT GET IT because of the coronavirus pandemic?"
- Spillovers to children:
  "At any time in the last 12 months, did any children in the household miss, delay or skip any PREVENTIVE checkups because of the coronavirus pandemic?"

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Appendix: Descriptive statistics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9	(10)	(11)
		Race/Ethnicity			Educational Attainment				Household Income		
	Total	White	Non-White	<HS	Some College	University	Post University	<50K	50K-100K	>100K	Missing Income
					Wave 13-21 (Pre-Policy)						
Panel A: Delayed Medical Care for Any Medical Condition											
Mean	0.342	0.333	0.370	0.293	0.353	0.337	0.357	0.376	0.342	0.321	0.316
S.D.	(0.474)	(0.471)	(0.483)	(0.455)	(0.478)	(0.473)	(0.479)	(0.484)	(0.474)	(0.467)	(0.465)
Panel B: Delayed Medical Care for Something Other Than COVID-19											
Mean	0.241	0.230	0.278	0.237	0.272	0.223	0.229	0.296	0.242	0.202	0.227
S.D.	(0.428)	(0.421)	(0.448)	(0.425)	(0.445)	(0.416)	(0.420)	(0.457)	(0.428)	(0.401)	(0.419)
N	694,961	529,389	165,572	921,23	222,913	203,662	176,263	176,540	195,517	233,456	89,448
					Wave 22-33 (Post-Policy)						
Panel D: Delayed Medical Care for Any Medical Condition											
Mean	0.231	0.220	0.269	0.206	0.245	0.226	0.233	0.275	0.229	0.205	0.209
S.D.	(0.422)	(0.414)	(0.443)	(0.405)	(0.430)	(0.418)	(0.423)	(0.446)	(0.420)	(0.404)	(0.407)
Panel E: Delayed Medical Care for Something Other Than COVID-19											
Mean	0.172	0.161	0.209	0.177	0.200	0.155	0.155	0.227	0.171	0.134	0.156
S.D.	(0.377)	(0.367)	(0.406)	(0.381)	(0.400)	(0.362)	(0.362)	(0.419)	(0.376)	(0.341)	(0.363)
Panel G: Vaccination Status (Received Vaccine)											
Mean	0.578	0.589	0.543	0.458	0.524	0.595	0.677	0.499	0.581	0.629	0.611
S.D.	(0.494)	(0.492)	(0.498)	(0.498)	(0.499)	(0.491)	(0.468)	(0.500)	(0.493)	(0.483)	(0.488)
N	700,070	536,820	163,250	85,746	219,559	207,707	187,058	189,943	209,140	248,919	52,068

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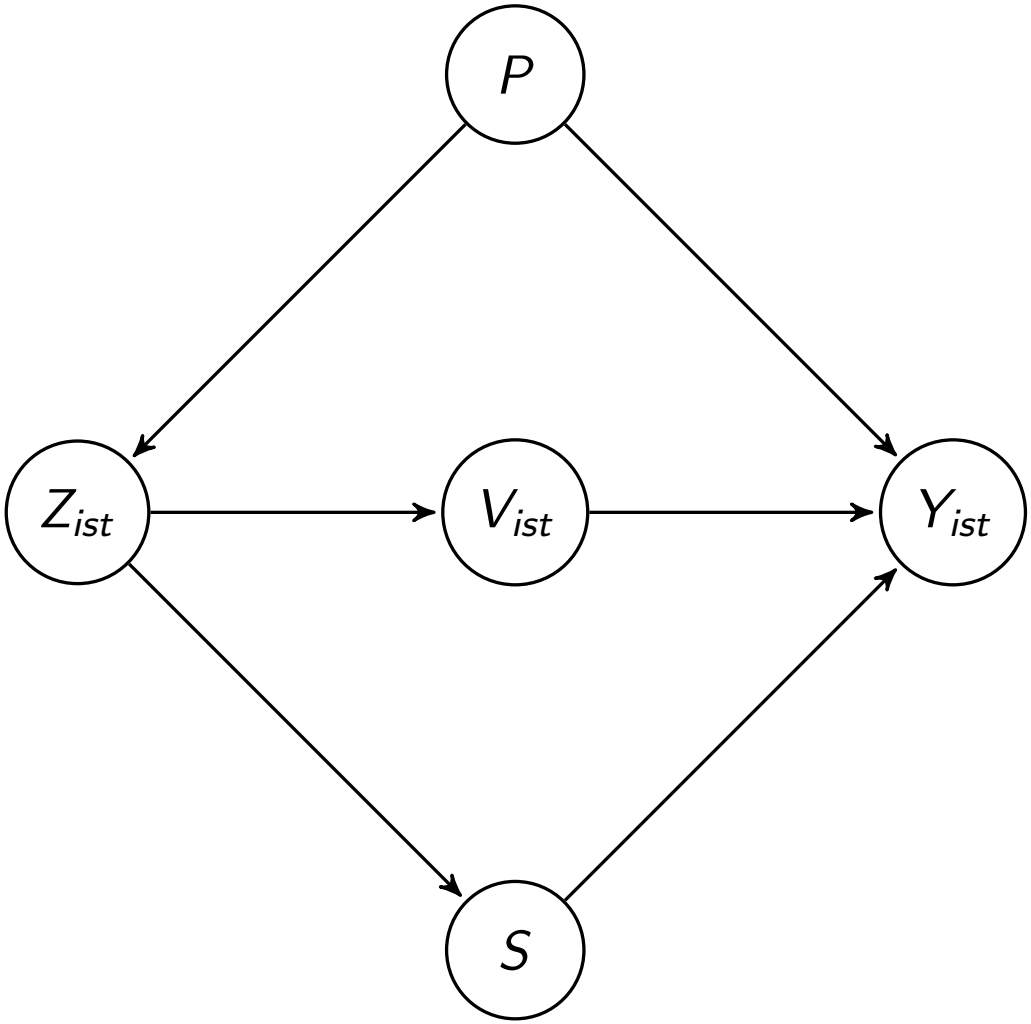
Appendix: A simple illustration of backdoor paths



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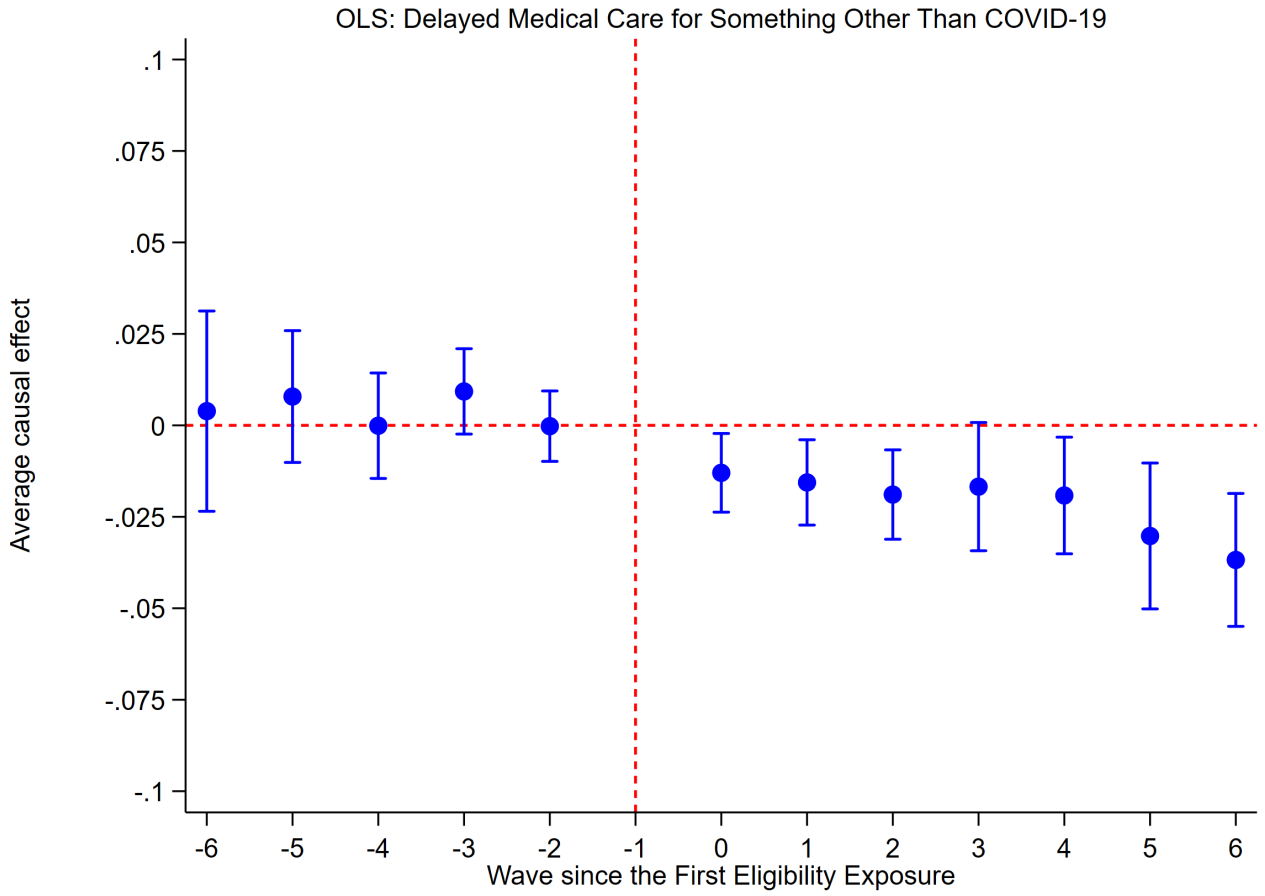
Appendix: dynamic DID - OLS



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Appendix: Trend analysis

	(1)	(2)	(3)	(4)	(5)	(6)
		OLS			IV	
Panel A: Delayed Medical Care for Any Medical Condition						
Received Vaccine	-0.0159***	-0.0159***	-0.0161***	-0.0608**	-0.0598**	-0.0540*
	(0.003)	(0.003)	(0.003)	(0.027)	(0.027)	(0.027)
Panel B: Delayed Medical Care for Something Other Than COVID-19						
Received Vaccine	-0.0217***	-0.0217***	-0.0218***	-0.0793***	-0.0793***	-0.0763***
	(0.003)	(0.003)	(0.003)	(0.024)	(0.023)	(0.027)
				First Stage Estimates (*Received Vaccine*)		
Vaccine Eligibility				0.1537***	0.1541***	0.1739***
				(0.011)	(0.012)	(0.014)
N	700,070	700,070	700,070	700,070	700,070	700,070
State FE	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√
Region Wave Linear Trend	√			√		
State Wave Linear Trend		√			√	
State × Wave FE			√			√
Kleibergen-Paap rk LM statistic				18.20	17.96	17.13
Kleibergen-Paap rk Wald F statistic				188.7	178.6	159.4

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Appendix: Feedback effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Delayed Medical Care for Any Medical Condition										
Vaccine Eligibility	-0.0133**	-0.0163**	-0.0164**	-0.0164**	-0.0164**	-0.0097**	-0.0111**	-0.0111**	-0.0111**	-0.0111**
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Vaccine Eligibility *(t+1)*	0.0050	0.0079	0.0079	0.0079	0.0079					
	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)					
Vaccine Eligibility *(t+2)*						0.0004	0.0016	0.0015	0.0015	0.0015
						(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
Panel B: Delayed Medical Care for Something Other Than COVID-19										
Vaccine Eligibility	-0.0115*	-0.0121*	-0.0118*	-0.0118*	-0.0118*	-0.0149***	-0.0158***	-0.0156***	-0.0156***	-0.0156***
	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Vaccine Eligibility *(t+1)*	-0.0007	-0.0003	-0.0005	-0.0006	-0.0006					
	(0.007)	(0.006)	(0.006)	(0.006)	(0.006)					
Vaccine Eligibility *(t+2)*						0.0044	0.0052	0.0050	0.0050	0.0050
						(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
N	700,070	700,070	700,070	700,070	700,070	700,070	700,070	700,070	700,070	700,070
State FE	√	√	√	√	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√	√	√	√	√
Region × Wave FE		√	√	√	√		√	√	√	√
Health measures and policy responses			√	√	√			√	√	√
Economic conditions				√	√				√	√
Social networks					√					√

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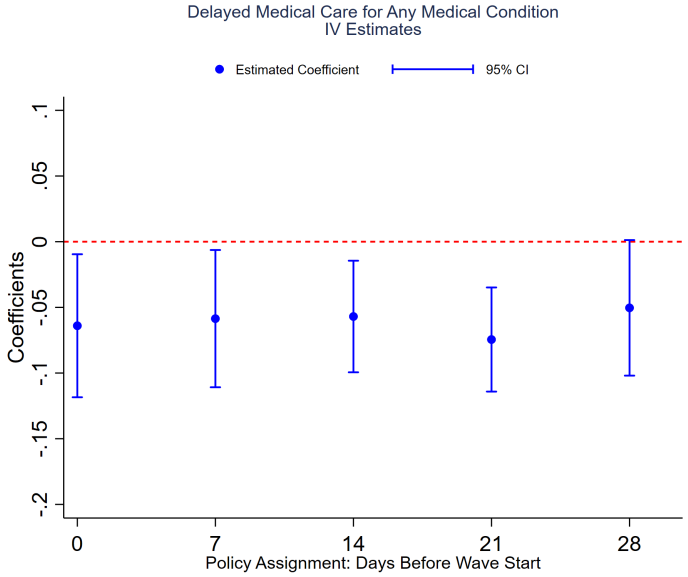
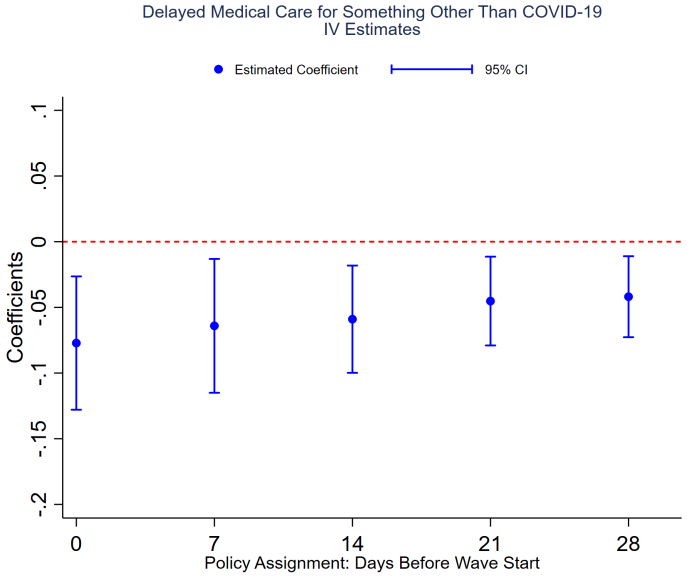
Appendix: Adjusting policy assignment





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Appendix: Other Expenditure Categories

	(1)	(2)	(3)	(4)	(5)	(6)
			IV			
	Food Service	Entertainment	Merchandise	Grocery	Transportation	All Spending
Vaccination Rate per 100	0.0004	0.0020	0.0057	-0.0014	0.0093***	0.0032*
	(0.0020)	(0.0050)	(0.0044)	(0.0026)	(0.0034)	(0.0018)
N	8,050	8,050	8,050	8,050	8,050	8,050
State FE	√	√	√	√	√	√
Survey Wave FE	√	√	√	√	√	√
Region × Wave FE	√	√	√	√	√	√
Health measures and policy responses	√	√	√	√	√	√
Economic conditions	√	√	√	√	√	√
Social networks	√	√	√	√	√	√
Kleibergen-Paap rk LM statistic	10.46	10.46	10.46	10.46	10.46	10.46
Kleibergen-Paap rk Wald F statistic	20.90	20.90	20.90	20.90	20.90	20.90

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